Department of Physics, Stanford University
PH 200-201, Advanced Physics Laboratory

EXPERIMENT S-3B (Rev. 1991)

HALL EFFECT (SEMICONDUCTOR)

OBJECT: To examine the Hall effect in a semiconductor.

GRNBRAL REFERENCES:

A. C. Melissinos, Experiments in Modern Phvsics, Sec. 3.3.

E. M. Purcell, Electricitv and Maanetism. Berkeley Phvsics Course. Vol. 2, Sec. 10.11.

A. F. Kip, Fundamentals of Electricity and Magnetism, 1969 Ch.9, QC519.K5

E. H. Putney, The Ball Effect and Related Phenomena, Ch. 1. chemistry QC612.H3.P8 .960

edited by C.L.Chien and C. R. Westgate, 1980, The Hall Effect and its Applications

Kittel, Charles 7th ed., 1996

INTRODUCTION:

In 1879, E. H. Hall observed that a magnetic field applied at right angles to the current flow in a conductor causes an electric field to be generated in a direction perpendicular to both.'" Although the basic explanation of this effect in terms of the ("Lorentz") force acting on the drifting charge carries is elementary[2] the subject has developed into a sophisticated part of solid-state physics.[3-5]

In the case of semiconductors, the Hall effect permits a determination of the dominant carrier of electricity, as well as more detailed properties such as the number of charge carriers per unit volume and their mobility. A good background is presented in Melissinos, Sec. 3.3, which must be studied carefully in order to understand the present experiment.

THEORY

Since this experiment deals with material in a magnetic field, a summary of pertinent formulas is given.[6] Because MKS units are now aooepted in Hall effect studies,[3] all equations will be in MKS units. But, for easier comparison with the extensive former literature, the same equations using the

---

* References are given at the end of this writeup,

Gaussian CGS system of units are shown in brackets.[7]

(1) Ordinary Hall Effect

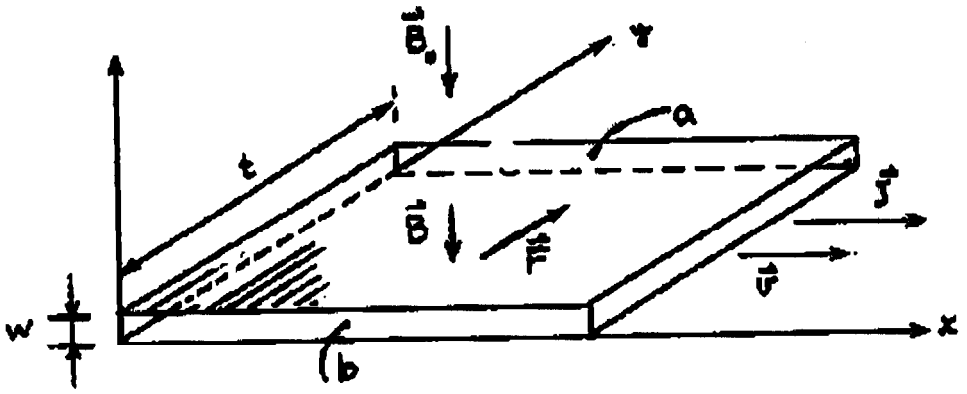


Fig. 1

If a current I of density $J_x = I/(wt)$ flows through the metallic strip shown in Fig. 1, the magnetic induction $\vec{B}$ will cause a force $\vec{F}$ on the current carriers of charge q and drift velocity $\vec{v}$:

$$\vec{F} = q\,\vec{v} + \vec{B} \qquad [\vec{F} = (q/c)\ \vec{v}\times\vec{b}] \tag{1}$$

Since

$$\vec{J} = n\,q\,\vec{v} \qquad [\ \vec{J} = n\,q\,\vec{v}\ ]\ , \tag{2}$$

where n is the number of charge carriers per unit volume, one can write Eq.(1)

$$\vec{F} = \frac{\vec{J}\times\vec{B}}{n} \qquad [\ \vec{F} = \frac{\vec{J}\times\vec{B}}{nc}\ ] \tag{3}$$

This shows, that independent of the sign of the charge, the charge carriers will be pushed against face a if the current flows in the + x direction, If the charge carriers are

electrons ($q = -e$), positive charge will accumulate on face b and negative charge on face a, setting up an electric field $\vec{E}_H$ in the + y direction. This field will exert an additional force

$$\vec{F}_H = -e\,\vec{E}_H \qquad [\ \vec{F}_H = -e\,\vec{E}_H\ ] \tag{4}$$

on each electron in the -y direction. Equilibrium will be reached when E_H has grown to a point where $\vec{F}_H + \vec{F} = 0$. Combining Eqs.(3) and (4), this condition yields

$$E_H = -\left(\frac{1}{ne}\right) J_x B \qquad [\ E_H = -\left(\frac{1}{nec}\right) J_x B\ ], \tag{5}$$

where account has been taken of the fact that $\vec{B}$ points in the -z direction. If the charge carriers are positive, the direction of $\vec{E}_H$ is reversed. If charge carriers of both signs are present, the above equations have to be modified.[8]

In Eq(5), the quantity E_H/J_x is called the Hall resistivity ρ_H in analogy with the ordinary resistivity E_x/J_x, where E_x is the field applied to the conductor to make the current I flow in the + x direction. According to Eq.(5),

$$\rho_H = E_H/J_x = R_H B \qquad [\ \rho_H = E_H/J_x = R_H B\ ], \tag{6}$$

where R_H is called the ordinary Hall coefficient:

$$R_H = -\frac{1}{ne} \qquad [\ R_H = -\frac{1}{ne}\ ]. \tag{7}$$

(2) Conductivity, Mobility

Figure 1 implies that there must be an externally applied electric field E_x which gives the charge carriers their drift velocity v_x in the x direction. The conductivity σ of the

material is defined by

$$\sigma = J_x/E_x \qquad [\ \sigma = J_x/E_x\ ] \tag{8}$$

and the mobility μ is defined as

$$\mu = v_x/E_x \qquad [\ \mu = v_x/E_x\ ]. \tag{9}$$

Using Eq.(2), Eqs.(8) and (9) lead to

$$\sigma = q\, n\, \mu \ . \tag{10}$$

From Eq.(7) it then follows (q = -e !)

$$\mu = R_H\, \sigma \qquad [\ \mu = R_H\, \sigma\ ] \tag{11}$$

Hence, the Hall effect can be used to determine the mobility of charge carriers.

(3) Semiconductors

In a semiconductor, there are positive charge carriers [holes in a filled band - see Melissinos, Fig. 3,19(b)], as well as negative charge carriers [electrons in the conduction band - see Melissinos, Fig. 3.19(a)]. Each of these charge carriers has a different drift velocity or mobility, Hence, Eq. (10) is replaced by

$$\sigma = e\,(n_h\mu_h + n_e\mu_e) \ . \tag{12}$$

Melissinos (p.86-87) shows that Eq.(ll) is replaced by

$$\mu_h^2 n_h - \mu_e^2 n_e = R_H\, \sigma^2 / e \ . \tag{13}$$

In Eqs.(12) and (13), the subscript h refers to holes and e to electrons. Equation (13) shows that under certain conditions R_H can be zero. If the semiconductor is of the "p" type, i.e., hole conductivity exceeds electron conductivity at room temperature, then, as the temperature is increased, n_e can increase more rapidly than n_h and R_H can change sign.

Melissinos (p.87-88) shows that a measurement of the conductivity at the inversion point (where $R_H = 0$) can lead to a determination of the mobility ratio μ_e/μ_h.

EQUIPMENT

The semiconductor crystal is mounted on a copper rod which can be heated or cooled in a manner similar to that shown in Melissinos, Fig. 3.21. The electrical connections to the crystal are similar to those shown in Melissinos Figs. 3.23 and 3.24 (but there is no contact 5). The basic electrical schematic is shown in Fig. 2 (not to scale!). There is also an electromagnet and its power supply. A copper constantan thermocouple[9] is attached to the base on which the crystal is mounted. The temperature of the base can read out directly on a calibrated meter.

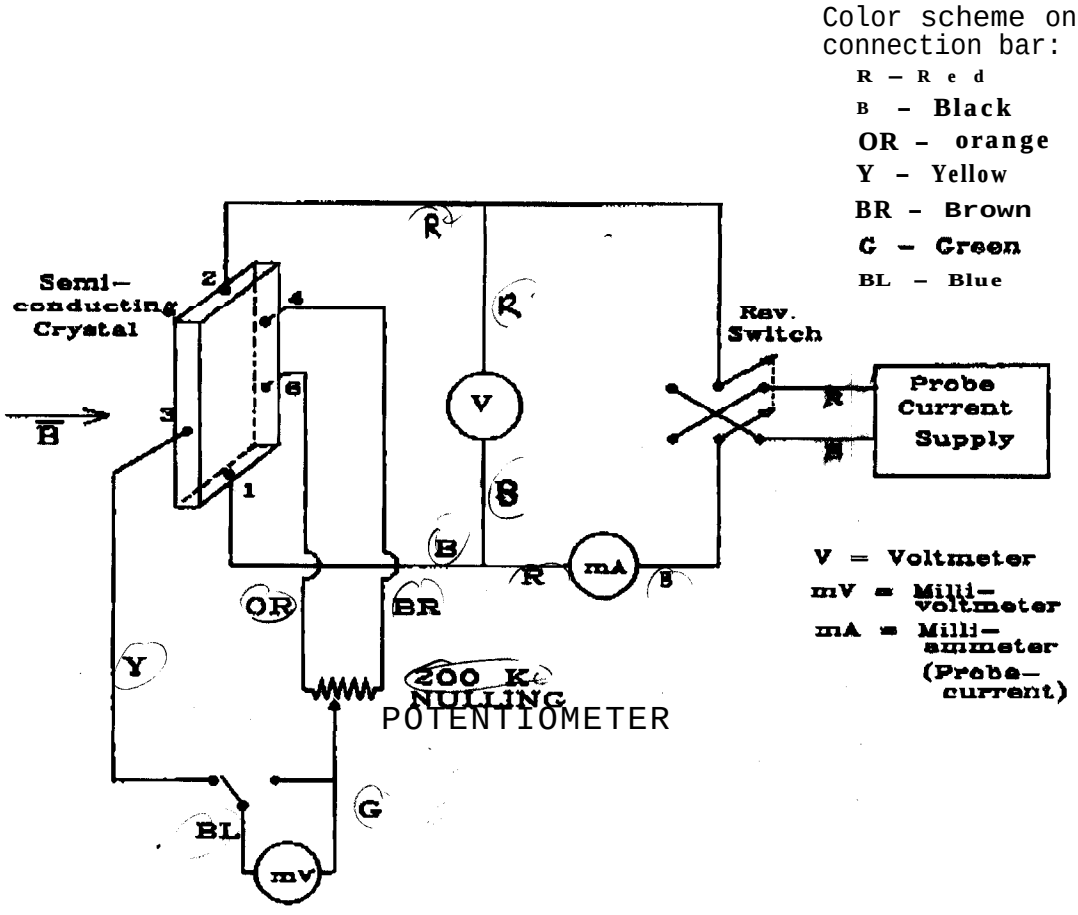


Fig. 2. Basic schematic for Hall effect measurement. Heater supply and magnet supply are not shown.

PROCEDURE

The procedure is very similar to that described in Melissinos, Sec. 3.3.4.

1) Set the voltage across the semiconductor sample so that a current of 5mA flows through the sample. Since some net current rectification of the contacts 1 and 2 on the sample (Fig.2) is unavoidable, use the reversing switch so as to select the current sense in which the smallest voltage is applied for a given current through the sample. Stay in this mode throughout the experiment.

2) The millivoltmeter needs approximately a 5 minute warm-up time before it stabilizes. Adjust the nulling potentiometer so as to produce a zero voltage across the millivoltmeter.' It is important to check this periodically, because of voltage drifts.

3) Apply a magnetic field of -1000 gauss across the sample and note the Hall voltage. A calibration curve for the magnetic field at the center of the gap is in the experiment file. As the coil of the electromagnet warms up, the voltage'of the magnet supply has to be readjusted to keep the current, i.e., magnetic field, constant. Reverse the magnetic field by first turning the magnet current to zero and then flipping the reversing switch on the supply. **IF YOU DO NOT DO THIS YOU WILL BURN OUT THE REVERSING SWITCH.** (why?) Is the magnitude of the Hall voltage independent of the direction of the magnetic field?

4) Using a straight wire, voltmeter, and current source, determine the direction of $\vec{B}$ in your setup. Use this to figure out the sign of the charge carriers in your sample at room temperature, i.e., whether the sample is n-type or p-type.

5) Check whether the Hall voltage is proportional to the sample current (use ~2,4,6,8 mA) and to the magnetic field (use -500, 1000, 1500 gauss).

6) **At some convenient magnetic field and sample current,** measure the Hall voltage and voltage across the sample (contacts 1 and 2 on Fig.2) as a function of temperature at a constant sample current (this differs from the procedure in Melissinos). DO NOT EXCEED + 90° C because the heater coil will. start to smoke (this limit corresponds to a heater voltage of ~20V or heater current of -0.2A). To obtain temperatures below room temperature, dip various lengths of the copper rod on which the sample is mounted into a Dewar filled with liquid nitrogen. At each **temperature point,** record the items (a), (b), (d) and (e) noted on top of p.92 in **Melissinos,** but replace "Resistivity" by sample voltage, measured only with the magnetic field off.

REPORT

1) Follow the data analysis in Melissinos, Sec. 3.3.5.

2) Make plot of the temperature dependence of resistivity in the intrinsic regibn and extract the gap energy.

3) Make plot to obtain the temperature dependence of the mobility in the extrinsic region.

4) Plot the "Hall mobility" **as** in **Melissinos,** Fig. **3.27.**

5) If your semiconductor sample has an inversion of the Hall voltage at a certain temperature, obtain the ratio of the hole to electron mobilities

6) **Obtain an order** of magnitude for the number of impurity carriers.

7) Discuss the overall behavior of your semiconductor **sample, i.e.,** does it have the expected overall characteristics?

REFERENCES

1. A fascinating personal account of Hall's discovery (which formed the subject of his Ph.D. thesis) can be found in Chien and Westgate, p. 523 ff.

2. D. Halliday and R. Resnick, Physics, Sec. 33-5.

3. See Chien and Westgate.

4. E. Bergmann, Physics Today, August 1979, p. 25 (reprint in experiment file.)

5. See Putney.

6. For derivations, see Kip, Ch. 9.

7. For a comparison of the unit systems, see Purcell, p. 449-452.

8. Melissinos, p. 86.

9. Melissinos, p, 105-106 and Figs. 1.8 and 3.22.